Noise and the fractal convergence boundary of gradient descent: first experiments

Arav Rawat · October 2026 · Follow-up to Liang & Montúfar [1] · Code and data: github.com/lnxv/fractal-convergence-noise

Question

Liang and Montúfar [1] show that gradient descent (GD) with near-critical step size on regularized scalar factorization has a fractal convergence boundary. Does a small amount of stochasticity preserve this boundary, blur it, or change it qualitatively, and how does the effect scale with the noise level?

Setup

$L(u,v) = ½(uv - y)^2 + ½\lambda(u^2 + v^2)$ with $y = 0.5$, $\lambda = 0.2$, $\eta = 1$, the setting of Figure 3 in [1]. On a 401×401 grid over $[-4,4]^2$, deterministic GD gives 126,880 escaping initializations and 33,802 near a minimizer after 100 steps, reproducing the region in [1, Fig. 3 middle]. Two noise models: (A) random step size $\eta_t = \eta(1 + \sigma\xi_t)$, $\xi_t \sim U(-1,1)$, applied to the full gradient; (B) additive gradient noise $\nabla L + \sigma Z_t$, $Z_t \sim N(0, I)$, a stylized proxy for SGD rather than minibatch sampling.

Method

With noise, each initialization has a *probability* of not escaping. A single noisy run per pixel is misleading: with independent draws at neighbouring pixels the picture becomes speckled (interface fraction 0.059 at σ = 0.1, versus 0.016 when all pixels share one noise path), so apparent roughening partly reflects sampling rather than geometry (Fig. 2, bottom). Following the probability-basin approach of Nieto, Seoane and Sanjuán [3], I estimate the survival probability $p(u_0,v_0)$ from 128 independent runs per initialization (horizon 200 steps; escape means |u| or |v| > 10^8) and measure the fraction of initializations with $0.1 < p < 0.9$, which scales as σ^β with β the noise-sensitivity exponent of [3]. For deterministic GD I measure the uncertainty exponent of Grebogi, McDonald, Ott and Yorke [2]: the fraction of initializations whose outcome flips under a perturbation of size ε scales as ε^{2-D}, where D is the box-counting dimension of the boundary. Experiments use two windows near the boundary, W1 = [2, 2.7]×[−0.3, 0.3] and W2 = [1.7, 2.3]×[0.5, 1.1], on 121×121 grids.

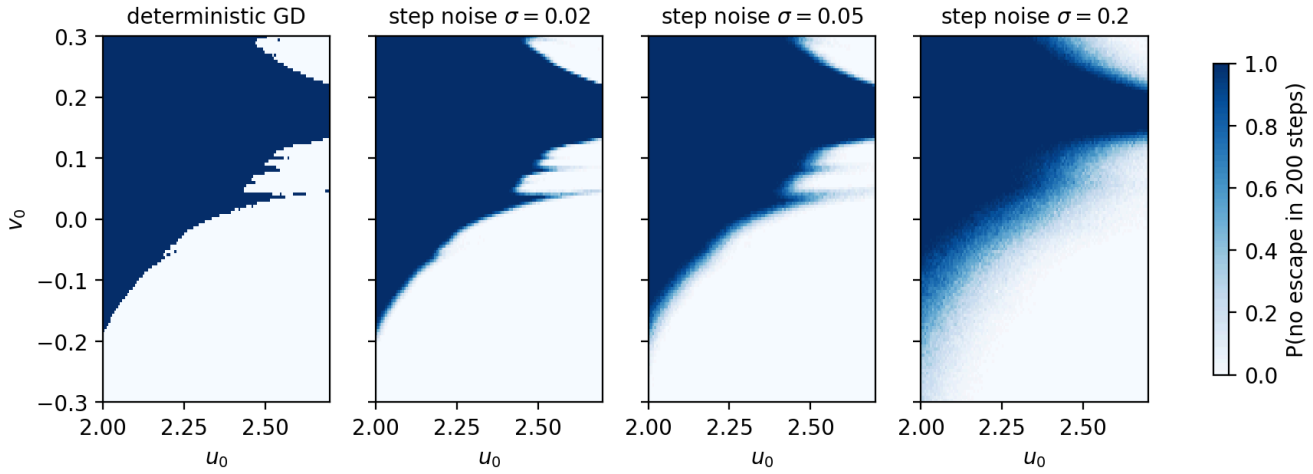


Figure 1. Survival probability in W1 for deterministic GD and step noise of increasing σ (128 runs per pixel).

Findings

- **Small noise blurs the boundary rather than preserving it.** The fractal interface becomes a band of uncertain outcomes (Fig. 1). For moderate σ the p = ½ contour is smoother than the deterministic boundary (interface fraction 20–35% lower), i.e. filaments thinner than the band are washed out.

- **The boundary barely moves.** The p = ½ contour disagrees with deterministic GD on 0.1–3.7% of each window across all tested σ, and the overall survival fraction changes by less than 0.014.
- **The uncertain area follows a power law in σ** (Fig. 2, $R^2 \geq 0.999$). The estimate is stable under 2× grid refinement (0.134 vs 0.135 at σ = 0.1) and doubling the horizon (0.136). An independent re-implementation (different language and random number generator) gives exponents within 0.02.
- **Dimension of the deterministic boundary.** The uncertainty exponent gives $D \approx 1.26$ in both windows, an independent estimate consistent with the box-counting value 1.249 reported in [1] for the projected boundary.

Window	β, step noise	β, additive noise	2 − D (deterministic)	D
W1	0.86 ± 0.01	0.84 ± 0.02	0.74 ± 0.01	1.26
W2	0.73 ± 0.01	0.83 ± 0.01	0.74 ± 0.01	1.26

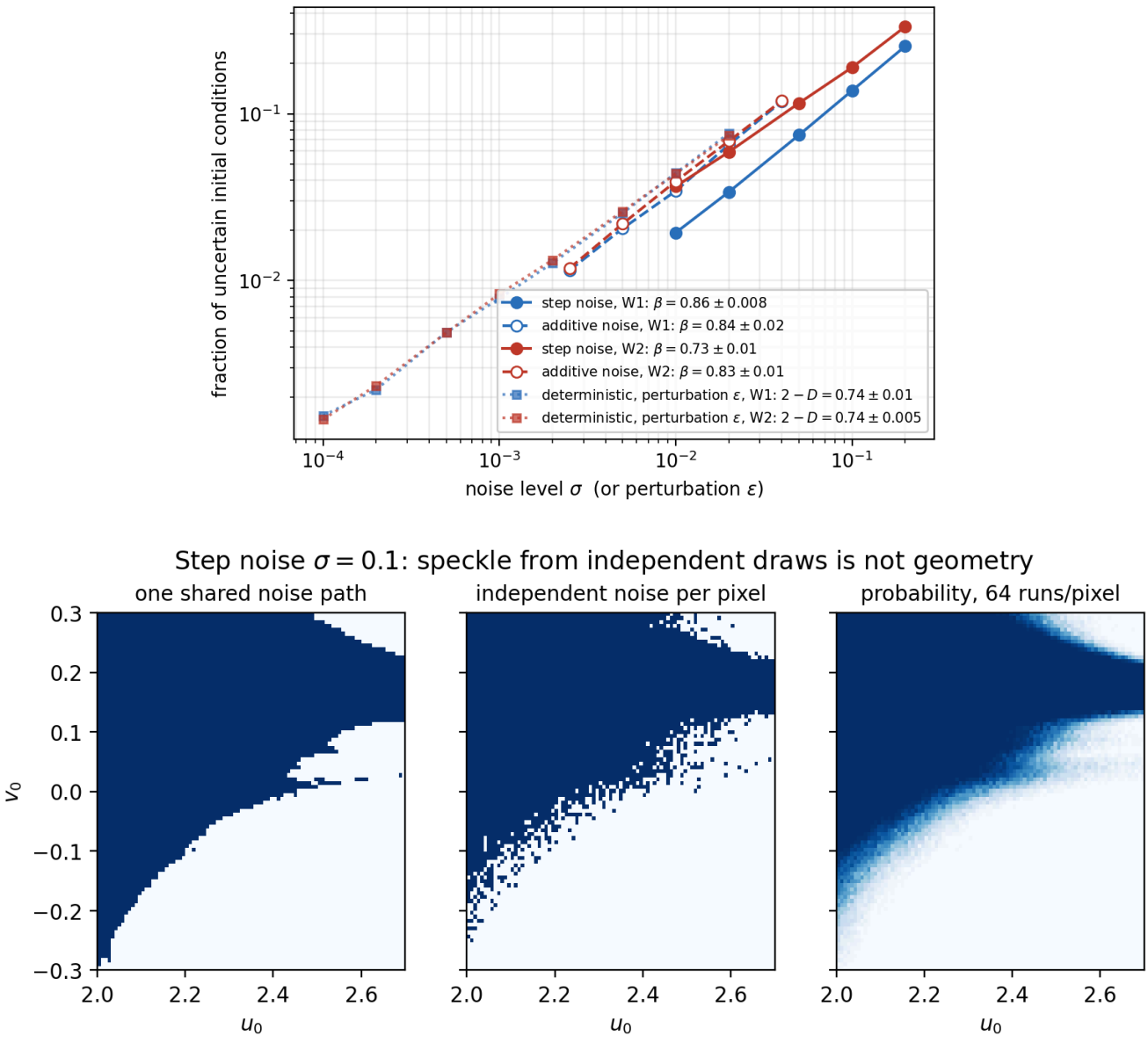


Figure 2. Top: fraction of uncertain initializations versus σ (circles) and versus perturbation size ε for deterministic GD (squares). Bottom: one shared noise path keeps a sharp boundary, independent draws per pixel produce speckle, and the probability map shows the actual band.

Interpretation and a possible mathematical question

If noise acted like a single random perturbation of the initialization with size proportional to σ, one would expect β = 2 − D. The noise exponents are of the same order but mostly larger (0.83–0.86 versus 0.74; step noise in W2 is the

exception at 0.73), so the effective perturbation seems to grow faster than linearly in σ near the boundary. One possible explanation is that noise injected at every step is amplified by the folding of the GD map described in [1]. A natural question is whether β can be derived from D and the expansion rate of the GD map on the boundary. Sohl-Dickstein [4] observed that fractal trainability boundaries in hyperparameter space survive minibatch noise; if the minibatch sequence is fixed across the grid, this corresponds to the shared-noise setting, where the boundary also stays sharp here (Fig. 2, bottom left).

Limitations

- Finite horizon: not escaping within 200 steps is not the same as asymptotic convergence, and only the escape/survive outcome is analysed, not which minimizer is selected.
- Two local windows only; the 0.1/0.9 thresholds are a choice; 128 runs per point (binomial standard error ≤ 0.044).
- Additive noise is a stand-in for SGD; true minibatch noise and matrix factorization remain to be tested.

References

1. S. Liang and G. Montúfar. Gradient descent with large step sizes: chaos and fractal convergence region. ICLR 2026. arXiv:2509.25351.
2. C. Grebogi, S. W. McDonald, E. Ott and J. A. Yorke. Final state sensitivity: an obstruction to predictability. Phys. Lett. A 99 (1983) 415–418.
3. A. R. Nieto, J. M. Seoane and M. A. F. Sanjuán. Final state sensitivity in noisy chaotic scattering. Chaos, Solitons & Fractals 150 (2021) 111181.
4. J. Sohl-Dickstein. The boundary of neural network trainability is fractal. arXiv:2402.06184 (2024).